Features and capabilities

Support Ticketing System — ITG Group

A support desk built for a garment and textile business: tickets are linked to the purchase orders, styles and shipments they actually concern, and service levels are measured against real working hours rather than the clock on the wall. Everything below is implemented and covered by automated tests. What is *not* included is listed at the end, in the same detail.

9 Ticket types	**10** Ticket states	**63** Permission keys	**7** Built-in roles
93 API endpoints	**55** Database tables	**21** Screens	**318 backend, 39 browser** Automated tests

Raising and working a ticket

Nine ticket types

Incident, Service Request, Software Bug, Data Correction, Access Request, Feature Request, Training Request, Security Incident and Integration Failure.

Ten states, with enforced transitions

New, Assigned, In Progress, Waiting for Requester, Waiting for Third Party, Escalated, Resolved, Closed, Reopened and Cancelled. Illegal moves are refused rather than silently allowed, so a ticket cannot skip from New to Closed.

Priority is calculated, never chosen

The requester describes impact and urgency — questions they can answer — and a configurable matrix produces the priority. Nobody is asked to pick a priority directly, because everybody picks the highest one available.

Requesters cannot declare their own emergency

A claim above the organization's cap is reduced and the original kept, so staff can see what the requester believed. Any lead can still raise it afterwards.

Conversation with internal notes

Public replies the requester sees, and staff-only notes they never do — excluded server-side rather than hidden by the interface.

Attachments, including screenshots and screen recordings

Files are identified by inspecting their contents, not by trusting the name, so a script renamed to .png cannot be served back as one.

Assignment, acceptance and hand-off

To a person or a team, with the previous owner, the reason and the method recorded.

Resolution and closure

A written resolution is mandatory. The requester confirms or rejects it, and a rejected resolution reopens the ticket rather than closing an argument.

Satisfaction rating

Overall, resolution and agent scores with an optional comment, once a ticket closes.

Service level agreements

Measured in business time, not wall clock

A four-hour target raised at 16:00 on Friday is not due at 20:00 that evening. Working hours, split shifts, half days and holidays are all honoured.

Working calendars

Any number of them, each with its own time zone, weekly pattern and holiday list. Recurring holidays repeat annually; moveable ones are entered per year.

Policies scoped by category, department and type

The most specific matching policy wins, with response and resolution targets per priority level.

A policy may set its own priority matrix

What counts as Critical is not the same question for a stopped production line as for an internal reporting request. Overrides are per cell, so a policy that decides two cells still inherits the rest.

Clocks that pause honestly

Time waiting on the requester or a third party can be excluded; time waiting on the support desk never is, because that is the delay the SLA exists to measure.

Warnings before the deadline, not only after

A configurable threshold raises a warning while there is still time to act.

Escalation and notifications

A configurable escalation ladder

Rungs fire at percentages of the resolution budget — warn at 70, chase the team lead at 90, again at 100 with a status change, department manager at 120. Each rung fires once, however often the monitor runs.

Breaches reach somebody even when nobody owns the ticket

An unassigned ticket that breaches is the most important breach there is, because nobody has looked at it. Supervision is notified regardless of assignment.

Interruption for the person who can act, a list for supervision

The assignee is shown a popup. Team leads, administrators and super admins get an entry in their notification list — interrupting them for every warning would teach them to dismiss all of them.

Email, when a mail server is configured

Every notification is queued for email as well as the application. Delivery is a queue drained by a worker rather than part of the request, so assigning a ticket never waits on a mail server, and a message that fails is retried with a widening gap before being given up on. A staging system can redirect every message to one address so a copy of production data cannot email real customers.

New work announces itself

Being assigned a ticket, or a ticket arriving in your team's queue, notifies everyone who has to pick it up. Nobody is notified of their own action.

Notifications cannot repeat themselves

Every notification carries a deduplication key, so a job that runs twice cannot tell the same person the same thing twice.

Knowledge base

Draft, review, published, archived

Each step is a separate permission. Nothing is visible to requesters until somebody entitled to publish has done so.

Full version history

Every save records a version with its author and a change note, so a reviewer can see what moved between two revisions.

Three visibility levels

Internal for staff only, Organization for requesters here, Public across tenants.

Suggestions while working a ticket

Articles matching the ticket's category and text are offered to the agent.

Helpfulness feedback

Readers mark an article helpful or not, so the useless ones become visible.

Reporting and audit

Five reports

SLA compliance, agent performance, over-claimed severity, ticket volume trend and satisfaction — each over a configurable period.

Staff workload, live

Open, in progress, waiting, Critical and High counts, SLA breaches, and the age of the oldest open ticket, per person. Administrators see the whole desk; a team lead sees only the teams they lead.

Dashboard

Open volume, breaches, unassigned queue depth and workload at a glance.

CSV export

Values that could be read as formulas are neutralised, so an exported report cannot execute anything when opened in a spreadsheet.

Append-only audit trail

Who did what, when, from which address, with the reason where one was required. Entries record the actor's name and email as a snapshot rather than a link, so a deleted account does not erase the history of what it did.

Administration

Users

Create, edit, deactivate, reset a password, revoke every session, and delete. Deleting an account that owns work anonymises it instead of destroying the history — the tickets survive and read “Deleted user”.

Roles and permissions

Sixty-three permission keys across eight areas. Roles are database rows, not hardcoded checks: nothing in the code branches on a role name, so a role the client invents works exactly like a built-in one.

Teams

Members, capacity weights, a lead, an escalation target and an acceptance timeout.

Service catalogue

Categories, subcategories, business applications and their modules, each able to route new tickets to a default team.

SLA policies and calendars

Described in full above.

System settings

Typed key and value pairs an administrator edits without a deployment.

AI assistance

Off by default and switchable per organization.

ERP integration

Tickets link to real business records

Purchase orders, styles, customers, suppliers, factories, merchants, production orders, inspections, shipments, invoices, debit notes, commission invoices, digital product passports and integrations.

Look a ticket up by the record it concerns

Support can answer “what is happening with this shipment” without knowing a ticket number.

Artificial intelligence, kept in its place

Optional, and off until switched on

Without a key the system behaves identically and every AI call reports itself unavailable.

It advises; the rules decide

The deterministic priority calculation always runs and always stands. A recommendation is shown beside it, never in place of it.

It cannot reach the database or bypass a permission

Suggestions are returned to a person, who acts on them under their own authority.

The key never leaves the server

It is not in the browser bundle, not in the database, and the interface has no code path that could reach a provider directly.

Security

Multi-tenant isolation, fail-closed

Every query is filtered by organization at the database layer. A missing filter returns nothing rather than everything.

Permission-based authorization throughout

Checked on the server for every action. A request that names an organization, a role or a permission is ignored — all three come from the signed-in token.

Data scopes

Own, assigned, team, department, organization or all. A ticket outside your scope answers “not found” rather than “forbidden”, so the list of what exists cannot be mapped by probing.

Passwords

PBKDF2-HMAC-SHA512 with a per-password salt. Lockout after repeated failures, and sign-in is rate limited per address.

Two-factor authentication

Time-based codes, per account.

Sessions

Short-lived access tokens with rotating refresh tokens. Reuse of a retired token is detected and recorded.

Content Security Policy and security headers

On both the API and the page the browser loads, verified in a real browser.

Ticket and article text is never rendered as markup

Which is what stops anybody who can write a comment running code in a colleague's browser.

SQL injection is structurally absent

One parameterised statement exists in the entire codebase; everything else goes through the query builder.

Refuses to start when misconfigured

A placeholder signing key, a wildcard CORS origin or demo seeding left switched on stops the application booting in production rather than running unsafely.

What is not included

Listed in the same detail as everything else. A capability list a client cannot trust is worse than none, because the first thing they find missing puts the rest in doubt.

Time recorded against a ticket

The database and the interface expect it, but no endpoint exists, so staff cannot log hours worked.

Escalation policies have no administration screen

A sensible default ladder is created on installation and can be changed only in the database.

Contact phone and tags are captured but never shown

A requester can supply a phone number and it is stored, but no screen displays it.

The workload figures have no screen yet

The data is available through the API; the page has not been built.

Staff are still called “agents” in places

A rename is planned and not yet done.

Notifications refresh every thirty seconds rather than instantly

Live push is installed but not yet connected.

Tickets cannot be raised by email

Every ticket is raised through the interface.

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Uploaded files are not scanned for malware

File types are verified and dangerous ones are never rendered in the browser, but no virus scanner is connected.

Container deployment is not yet working

The API image builds; the browser application image does not. Installation is manual for now.

Email is the one to weigh first. Everything else on this list is a convenience or an internal tool. A support desk whose requesters are never emailed expects them to log in and look, which changes how the desk is used.

How it is built

ASP.NET Core 10 and C# on the server, React on the browser, SQL Server for storage. The server is layered so that the business rules depend on nothing — not the database, not the web — and seven automated tests fail the build if that is ever violated. Three hundred and eighteen server tests and thirty-nine browser tests run on every change, the server tests against a real SQL Server rather than a substitute, because every isolation defect this project has found reproduced only against a database that genuinely applies the rules.

Counts in this document were taken from the repository and the database, not from memory. Regenerate with `python tools/generate-feature-guide.py`.